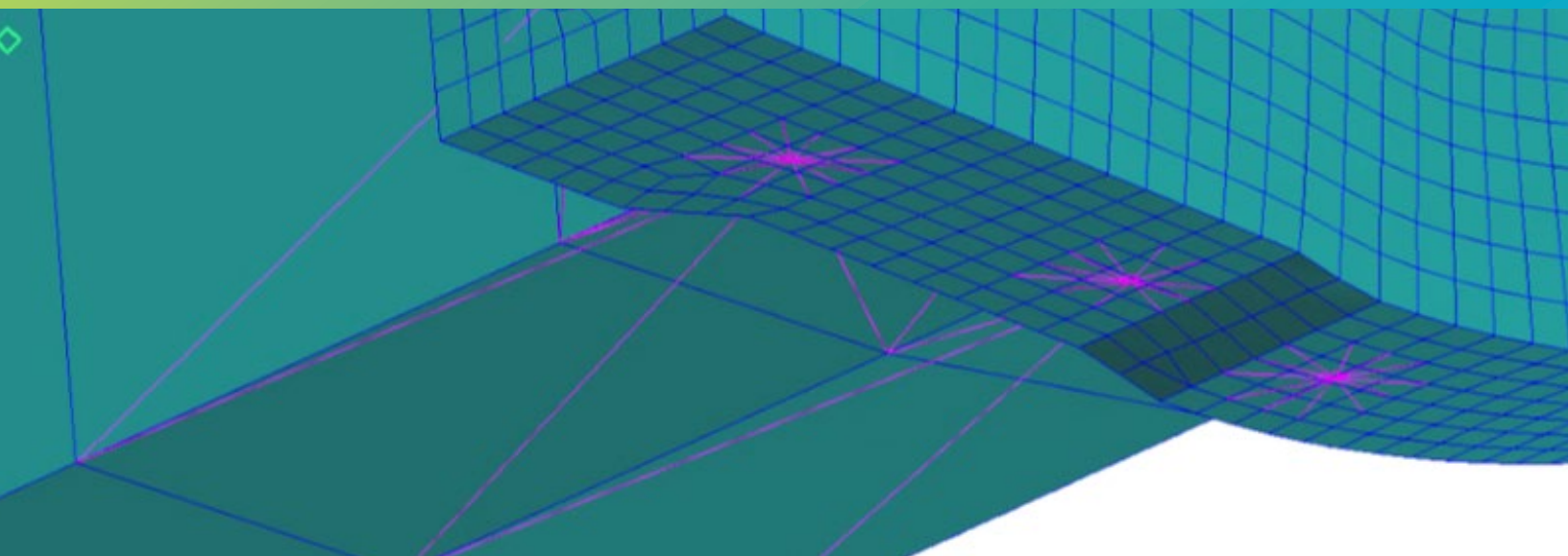


Gulfstream Aerospace

Gulfstream Aerospace cuts time and manpower required to model wing components in half by using MSC Nastran and Patran



Introduction

Aerodynamic structures need to be designed not only to resist static loads encountered in flight but also dynamic loads in order to avoid flutter, a potentially destructive phenomenon in which aerodynamic forces couple with a structure's natural mode of vibration.

For this reason, an important step in the design process is structurally analyzing the rudder aileron and spoiler attachments to the airframe to generate stiffness values that can be used by the dynamics group to evaluate aeroelastic performance. The process of modeling these attachment areas is complicated by the presence of large numbers of fasteners and the need to transition from the global model's coarse mesh over the expanse of the wing and control surfaces to the much finer mesh required in the attachment area.

Challenge

In the past, Gulfstream engineers modeled fasteners in the attachment areas using the Huth method, which required hand calculations in Excel spreadsheets to determine stiffness values. This was a tedious, manual process and the fastener nodes also required alignment with the mesh, which required a great deal of planning and manual mesh alignment. Each control surface attachment typically has about 75 fasteners, each of which takes about 5 minutes to analyze using the Huth method for a total of about 375 minutes per component.

Another difficulty with the previous modeling method was the need to manually create a transition zone between the fine mesh needed in the attachment area and the coarse mesh which exists in the original global model. Gulfstream engineers originally created the transition from coarse quadrilateral elements (CQUAD) to fine quads using triangular (CTRIA) elements, which required considerable manual effort and also reduced the accuracy of the analysis due to the inherent limitations of CTRIA elements. The CTRIA elements were needed to slowly transition between each large rectangle in the coarse area to over 40+ elements in the fine region.

These activities had to be performed on each control surface of each new aircraft design, typically the rudder, port and starboard elevators, port and starboard ailerons, and all six spoilers. The rudder has four hinge locations and two actuators, ailerons have two actuators and four hinges, the elevators have two actuators and five hinges, and the spoilers have one actuator and four hinges apiece. Modeling and analysis of these points needs to be completed in one to two months in order to meet the scheduling requirements for typical aircraft program.

“The large manual effort involved in the modeling process required a team consisting of six people for most of the process,” said Brian Smith, Stress Engineer for Gulfstream. “Another concern with the previous approach is the use of CTRIA elements. Usually, CQUAD elements are preferred over the triangular elements. The CTRIA element is a constant strain element. It is excessively stiff and it is generally less accurate than the CQUAD element. Whenever feasible, the CQUAD element should be used instead of the CTRIA element.”

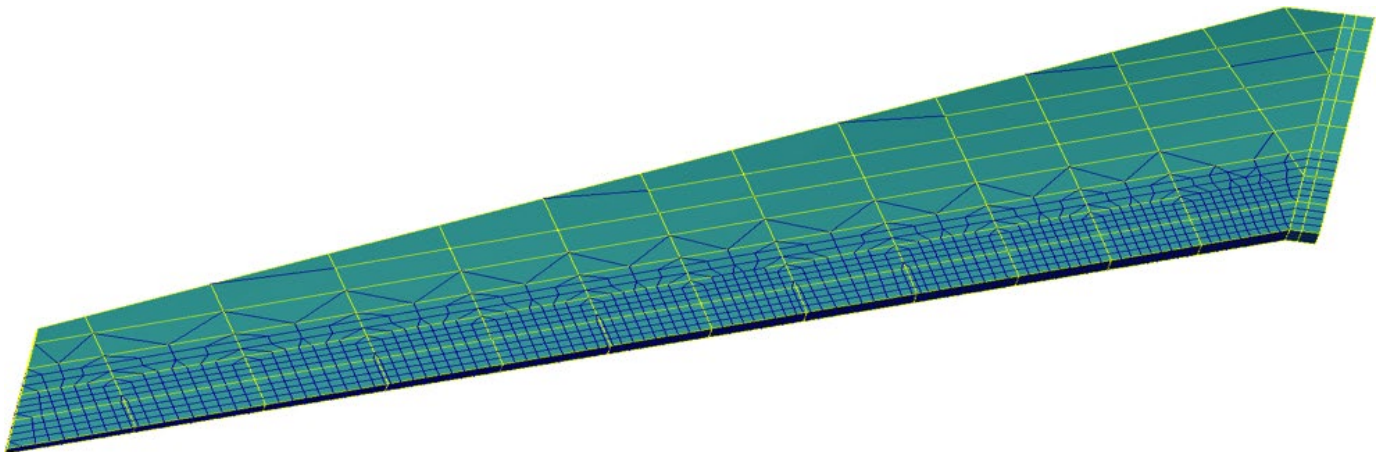


Fig. 1: Transition from coarse CQUADs to fine CQUADs with CTRIAs

Key highlights:

Product: MSC Nastran and Patran

Industry: Aerospace

Benefits:

- Time required to model a fastener on a wing reduced from over five hours to 30 seconds
- Number of engineers required to model components reduced by 2/3
- Improvements through MSC Nastran and Patran made it possible to use a lighter and less expensive structure



The new approach has substantially reduced the amount of time required to model fasteners and coarse-fine mesh transitions while also increasing accuracy of the model”

Rodrigo Guimaraes,
Stress Engineer for Gulfstream

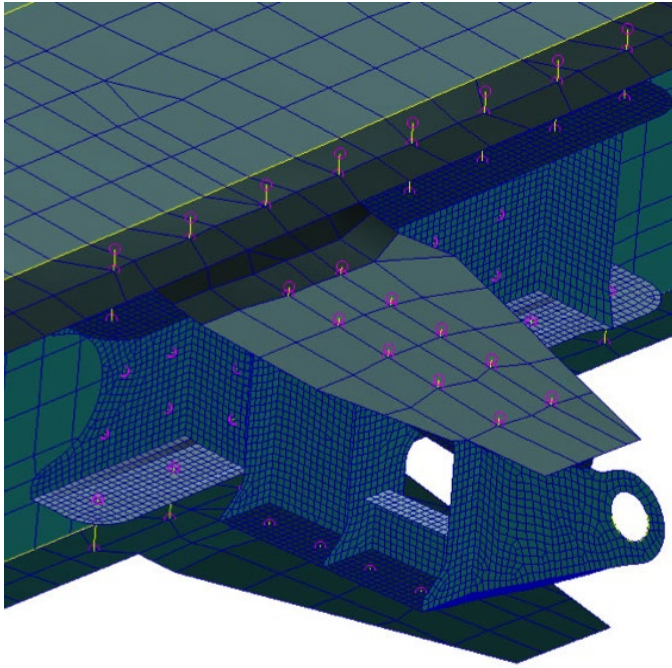


Fig 2: Rutman fastener utility automated process of modeling fasteners

Solution/validation

Gulfstream now uses the Rutman fastener utility embedded in Patran to automate the process of modeling fasteners. The Gulfstream engineer models the fastener simply by specifying the fastener material, diameter grip length, and a few other parameters in Patran. The utility then automatically models the fastener using a combination of CBAR, RBAR and CBUSH, which represents the complete work of a fastener joint. “The use of Rutman fasteners allows Gulfstream engineers to connect the backup structure with its actual geometry instead of an idealized and simplified representation, increasing the accuracy of the stiffness model,” said Rodrigo Guimaraes, Stress Engineer for Gulfstream.

Gulfstream engineers improved the process of transitioning between fine and coarse meshes by utilizing the MSC Nastran Edge-to-Edge glued contact to connect elements with different sizes. MSC Nastran searches from the slave node to the master node to detect the contact and once the contact is detected, MPC equations are generated to tie the slave node to the master nodes. This greatly simplifies the process of connecting dissimilar meshes and makes it possible to transition from coarse to fine meshes without transition elements. “By eliminating the need for a transition mesh, glued contacts make it possible to reduce the area where a fine mesh is required, which in turn reduces the number of elements by about 50% and provides a roughly proportionate reduction in the amount of time required to solve the model,” Smith said. The glued contact also eliminates the need for CTRIA elements and reduces remeshing requirements when the design changes.

Results

“The new approach has substantially reduced the amount of time required to model fasteners and coarse-fine mesh transitions while also increasing accuracy of the model,” Guimaraes concluded. The time to model each fastener has been reduced to 30 seconds, providing a time savings of 5 hours and 37 minutes per component. For a typical wing with six components, the time savings add up to 33 hours and 45 minutes for fasteners alone. Substantial time is also saved in transitioning between coarse and fine meshes. The time required to model each component has been reduced from two to four weeks to one week, which should make it possible in future projects with a maximum of only two engineers, one-third of the number required in the past. Finally, on a recent project, the higher mesh accuracy provided the ability to reduce conservatism in the analysis by more accurately representing the actual structure, thereby allowing for Gulfstream to take advantage of the structural capabilities previously not present in the FEA model. This led to an increase in downstream stiffness by 71% for the inboard elevator backup structure, and 62% for the outboard elevator backup structure. The refined mesh also provided for the backup structure’s torsional stiffness increase by about 61%. These improvements have made it possible to justify a lighter and less expensive structure.

About Gulfstream

Gulfstream Aerospace Corporation, a wholly owned subsidiary of General Dynamics designs, develops, manufactures, markets, services and supports advanced business-jet aircraft. Gulfstream has produced more than 2,000 aircraft for customers around the world since 1958. Gulfstream offers a comprehensive fleet of aircraft, comprising the wide-cabin, high-speed Gulfstream G150®; the new large-cabin, mid-range Gulfstream G280®; the large-cabin, long-range G450®; the large-cabin, ultra-long-range Gulfstream G550® and the ultra-large-cabin, ultra-long-range G650®. Gulfstream also offers aircraft ownership services via Gulfstream Pre-Owned Aircraft Sales®. The company employs more than 13,000 people at 12 major locations.

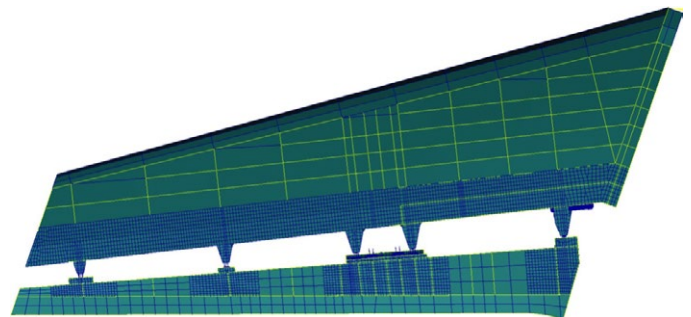


Fig. 3: The glued contact eliminates the need for the transition zone



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Our technologies are shaping production and people-related ecosystems to become increasingly connected and autonomous – ensuring a scalable, sustainable future.

Hexagon's Manufacturing Intelligence division provides solutions that use data from design and engineering, production and metrology to make manufacturing smarter. For more information, visit hexagonmi.com.

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